Traceable Predictive Maintenance for NASA C-MAPSS: An End-to-End PHM System with Calibrated RUL Prediction, Deterministic Risk Logic, and Interactive Scenario Analysis

CMAPSS-PHM Project

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Abstract

This paper presents an end-to-end predictive maintenance application built on the NASA C-MAPSS benchmark. The system combines a multi-model predictive layer for Remaining Useful Life (RUL) estimation, a deterministic agent layer for risk scoring and maintenance recommendations, and a dashboard layer for decision presentation, scenario analysis, and technical auditability. Candidate models include Random Forest, HistGradientBoosting, LSTM, and GRU. Although GRU achieved the best raw RMSE, the deployed champion is Random Forest because it offered stronger stability and integration readiness. The application augments point predictions with calibrated confidence bands, converts them into traceable risk and recommendation outputs, and exposes the full workflow through interactive user and technical-audit views. The result is both a scientific study on C-MAPSS PHM and an implemented AI system suitable for reproducible deployment-oriented evaluation.

1 Introduction

Remaining Useful Life (RUL) estimation is a core task in prognostics and health management (PHM), but benchmark-oriented studies often stop at predictive model scores. Real maintenance systems need uncertainty handling, deterministic decision logic, scenario analysis, explainability, and traceability in addition to point estimates. This work addresses that integration gap by implementing and evaluating a layered PHM stack over NASA C-MAPSS data [1, 4].

Contributions.

- A calibrated and benchmarked predictive layer with champion selection driven by accuracy and operational constraints.
- A deterministic agent layer that maps predictive outputs to auditable risk and recommendation decisions.
- A dashboard layer that operationalizes summary, analysis, scenarios, and technical-audit workflows.
- A reproducible artifact package linking data, code, models, contracts, and runtime traces.

2 Related Work

Research around C-MAPSS has heavily emphasized predictive performance, especially with sequence models such as LSTM and GRU [3, 6]. Tree-based methods remain strong tabular baselines for robustness and low-latency deployment. PHM deployment literature additionally highlights the importance of uncertainty communication, actionability, and governance in safety-critical settings [5, 2]. This paper positions itself between benchmark modeling and deployable PHM decision-support systems.

3 Dataset and Engine Context

3.1 NASA C-MAPSS problem setting

C-MAPSS represents multivariate run-to-failure trajectories generated from a turbofan simulation framework [1]. Each row contains a unit identifier, cycle index, three operational settings, and sensor measurements. The four benchmark subsets differ in operating-condition and fault-mode complexity, summarized in Table 1.

Table 1: CMAPSS subsets and problem structure.

Subset	Train units	Test units	Conditions	Fault modes
FD001	100	100	1 (sea level)	1 (HPC)
FD002	260	259	6	1 (HPC)
FD003	100	100	1 (sea level)	2 (HPC + Fan)
FD004	248	249	6	2 (HPC + Fan)

3.2 Engine schematic context

Figure 1 provides a conceptual turbofan degradation and monitoring view aligned with the NASA C-MAPSS guide. Operational settings and sensor channels define the context in which degradation patterns are observed cycle by cycle.

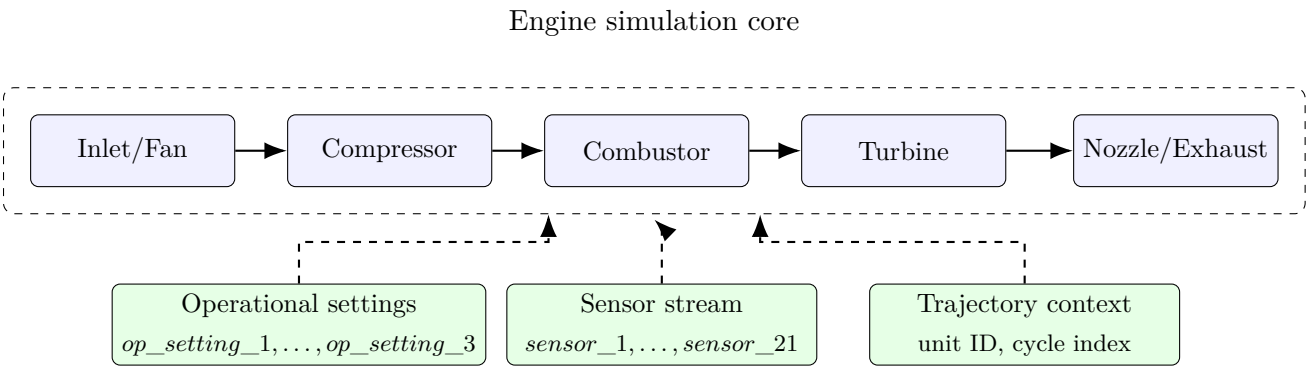


Figure 1: Conceptual C-MAPSS turbofan monitoring schematic, adapted from the NASA C-MAPSS user-guide context.

4 Methodology

4.1 Preprocessing and feature policy

Preprocessing removes low-variance variables, fits a training-only standard scaler, and preserves both raw and capped RUL policies. Table 2 summarizes the selected and removed feature sets used by the deployed predictive pipeline.

Table 2: Selected vs removed low-variance features.

Selected predictive features (16)	Removed low-variance features (8)
`op_setting_1`, `op_setting_2`, `sensor_2`, `sensor_3`, `sensor_4`, `sensor_7`, `sensor_8`, `sensor_9`, `sensor_11`, `sensor_12`, `sensor_13`, `sensor_14`, `sensor_15`, `sensor_17`, `sensor_20`, `sensor_21`	`op_setting_3`, `sensor_1`, `sensor_5`, `sensor_6`, `sensor_10`, `sensor_16`, `sensor_18`, `sensor_19`

4.2 Candidate model portfolio

Table **??** summarizes the candidate-model roles.

Table 3: Candidate models and intended roles.

Model family	Input shape	Core strengths	Role
Random Forest Regressor	Row tabular	Robust baseline, fast iteration, stable serving	Baseline / champion candidate
HistGradient Boosting Regressor	Row tabular	Competitive baseline, efficient training	Baseline / fallback candidate
LSTM	Windowed sequence	Temporal degradation modeling	Sequence challenger
GRU	Windowed sequence	Strong temporal pattern capture	Sequence challenger

4.3 Architecture and deterministic boundaries

Figure 2 shows the complete system flow and contracts.

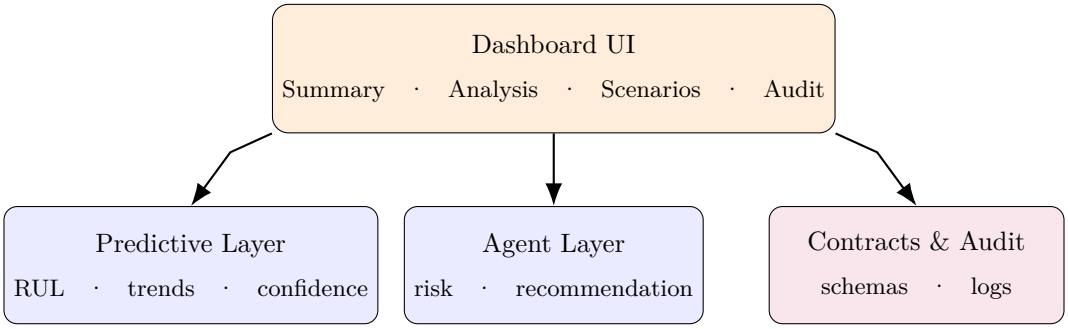


Figure 2: Simplified PHM architecture highlighting the dashboard as a multi-source consumer of predictive inference, agent-based decision logic, and audit infrastructure.

5 Experiments and Results

5.1 Global and per-dataset performance

Table 4 shows global metrics, while Table 5 details subset behavior.

Table 4: Global model metrics.

Model	RMSE	MAE	Rows
GRU	17.424	13.194	25,309
RF	18.313	13.256	32,267
HGB	18.433	13.329	32,267
LSTM	20.012	14.990	25,309

Table 5: Per-dataset RMSE by model.

Model	FD001	FD002	FD003	FD004
GRU	13.974	19.427	14.121	17.989
RF	18.686	19.600	16.978	17.512
HGB	18.680	19.763	17.065	17.654
LSTM	18.784	21.728	15.733	20.468

Table 6: Latency and engineering trade-offs.

Model	Samples	Total seconds	ms/sample
HGB	512	5.493	10.728
RF	512	0.161	0.313
LSTM	28,149	0.383	0.014
GRU	25,309	1.319	0.052

5.2 Champion decision

Although GRU obtains the best raw RMSE, the deployed champion remains RF due to stability and deterministic integration-readiness with an HGB fallback. This emphasizes application-level reliability over leaderboard-only optimization.

5.3 Scenario and reliability evidence

Table 7: Scenario comparison example.

Metric	Baseline	Scenario
RUL prediction	123.955	123.874
Risk level	healthy	healthy
Risk score	35	35
Service status	ok	ok
ΔRUL	-0.081	
ΔRisk score	0	

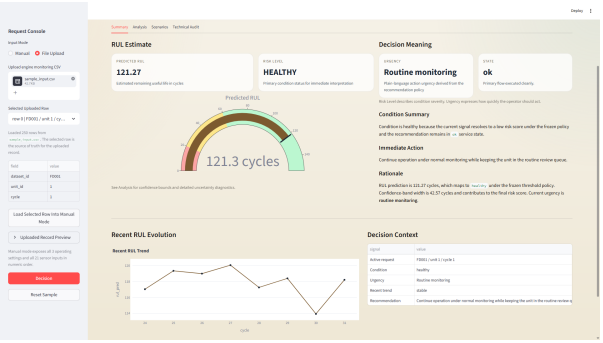
Table 8: Failure modes and handling.

Failure mode	Handling strategy
Invalid request payload	Deterministic validation response with explicit error message
Predictive validation failure	Propagate `error_validacion` service state
Champion unavailable	Use deterministic fallback model and state propagation
LLM key/provider failure	Continue with rules-only deterministic path
Unsafe scenario perturbation	Reject scenario with validation errors and safety notes

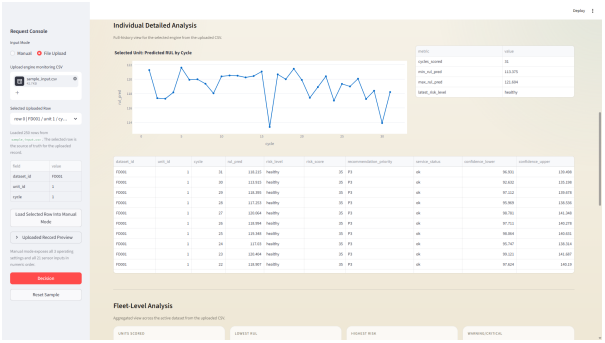
Table 9: Contract and trace field mapping.

Dashboard tab	Field	Source layer
Summary	`rul_pred`, `risk_level`	agent / predictive
Analysis	`risk_score`, `confidence_band`, `trace`	agent / predictive
Scenarios	`comparison`, `change_summary`	agent scenario
Technical Audit	`audit_record_id`, `model_version`, `service_status`	agent / predictive

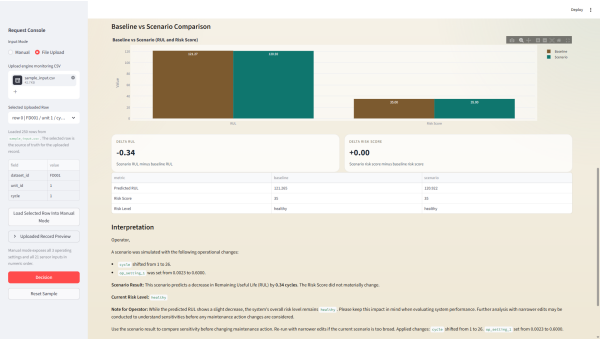
5.4 Dashboard evidence and visual results



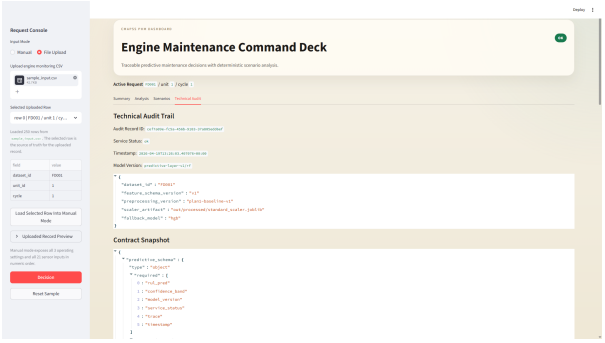
(a) Summary tab.



(b) Analysis tab.

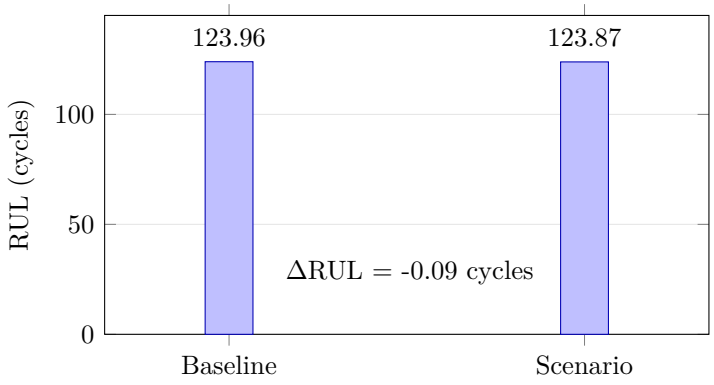


(c) Scenarios tab.

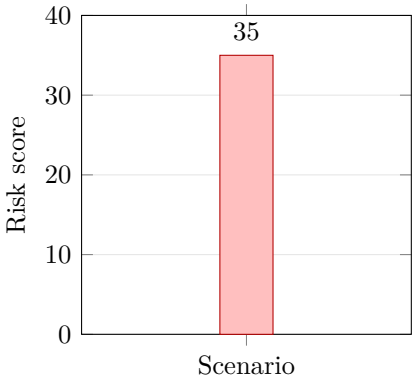


(d) Technical Audit tab.

Figure 3: Dashboard evidence across operator and audit workflows.



(a) Baseline-versus-scenario RUL comparison.



(b) Scenario risk score.

Figure 4: Scenario comparison split into physical prediction and decision-oriented risk: (a) baseline and scenario RUL estimates with the associated ΔRUL, and (b) the scenario risk score.

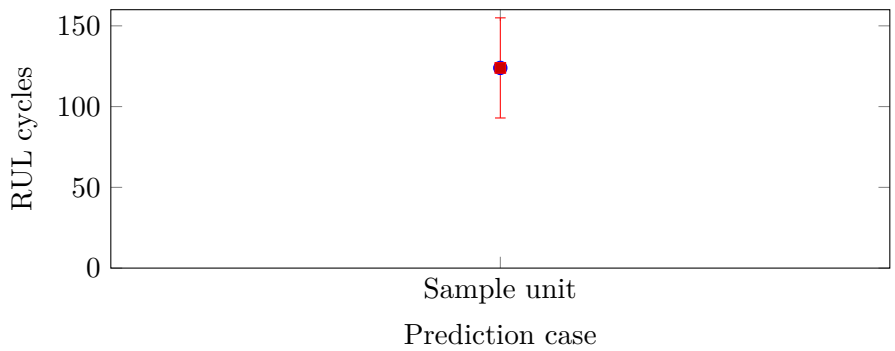


Figure 5: RUL interval illustration using calibrated confidence-band policy around a point prediction.

6 Discussion

The key engineering lesson is that the best benchmark model is not always the best deployed model. In this system, RF was selected over GRU to prioritize stable serving behavior, deterministic fallback, clear traceability, and low integration overhead. Scenario analysis further demonstrates that not all domain-meaningful perturbations produce large predictive effects if changes target excluded or locally insensitive features.

7 Conclusion

This work delivers a full PHM stack for C-MAPSS that integrates prediction, uncertainty, deterministic risk logic, scenario comparison, and auditability. Scientifically, it demonstrates calibrated RUL modeling with explicit champion-selection rationale. Engineering-wise, it demonstrates traceable end-to-end deployment with contract-driven boundaries and robust handling of degraded states.

References

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